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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 22.5

$$\begin{aligned}
 & \int x\sqrt{1-x^2} \operatorname{Bgsin}(x) dx \\
 & \left\{ \begin{array}{l} u = \operatorname{Bgsin}(x) \\ dv = x\sqrt{1-x^2} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \frac{dx}{\sqrt{1-x^2}} \\ v = -\frac{(1-x^2)^{\frac{3}{2}}}{3} \end{array} \right. \\
 & = -\frac{\operatorname{Bgsin}(x)(1-x^2)^{\frac{3}{2}}}{3} + \frac{1}{3} \int \frac{(1-x^2)^{\frac{3}{2}}}{\sqrt{1-x^2}} dx \\
 & = -\frac{\operatorname{Bgsin}(x)(1-x^2)^{\frac{3}{2}}}{3} + \frac{1}{3} \int (1-x^2) dx \\
 & = -\frac{\operatorname{Bgsin}(x)(1-x^2)^{\frac{3}{2}}}{3} + \frac{1}{3} \left(x - \frac{x^3}{3} \right) + C \\
 & = -\frac{\operatorname{Bgsin}(x)(1-x^2)^{\frac{3}{2}}}{3} + \frac{x}{3} - \frac{x^3}{9} + C
 \end{aligned}$$

References